Solution Requirements

HDI Prediction System | Team Lead: Farzana | Contributors: Chinthoti Raghavendra, Ahmad Kabir Shaik, Naga Sireesha Naralasetty, Sitha Ramaraju Velisala

Functional requirements

01 The system shall accept user input for key socio-economic indicators through a web form.

02 The system shall validate that entered indicator values are within a reasonable numeric range.

03 The system shall use a trained Linear Regression model to predict the HDI score from the inputs.

04 The system shall display the predicted HDI score to the user immediately after submission.

05 The system shall allow users to submit multiple predictions in a single session without reloading.

Non-functional requirements

01 Performance: prediction results should be returned within 1–2 seconds of submission.

02 Usability: the interface should be simple enough for non-technical users to operate without guidance.

03 Accuracy: the underlying model should maintain an R^2 score of at least 0.90 on test data.

04 Availability: the deployed application should be accessible via a public link at all times.

05 Maintainability: code should be modular, separating data processing, model, and web layers.

Data requirements

01 A cleaned 2021 global HDI dataset covering health, education, and income indicators for ~195 countries.

02 Consistent column naming between the training dataset and the web form input fields.

Constraints

01 Limited to indicators available in the 2021 dataset; no real-time data integration.

02 Deployment restricted to free-tier hosting (Render.com), which may have cold-start delays.

03 Academic project scope — not intended for official policy or certification use.